

Aurangzaib Shehzad Awan

GenAI (Agentic Systems)

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Career Objective

GenAI Engineer specializing in multi-agent systems, RAG pipelines, and production-ready LLM applications. Experienced in building and deploying intelligent systems using modern LLM frameworks and MLOps tooling.

Experience

GenAI Engineer Intern · Systems Limited

June 2026 – Present

- Built a multi-agent conversational system for a high-end fashion brand: a browser agent handling routine product and catalog queries, and a case-solver agent handling complex, high-caliber requests.
- Defined the routing logic between agents so query complexity determined which agent handled the request, keeping simple lookups cheap and reserving deeper reasoning for hard cases.
- Engineered a token-optimization strategy for prompt construction, cutting LLM token usage and inference cost, and introduced multithreading to improve pipeline throughput.

AI/ML Engineer Intern · Netsol

Jun 2025 – Aug 2025

- Built a web application with FastAPI and React, containerized it with Docker, and deployed it serverlessly on AWS Lambda.

Projects

MentorAI – FYP *FastAPI, React, YOLOv8, MediaPipe, PyTorch, LangGraph*

- Engineered a stateful AI agent using LangGraph and Gemini/Groq LLMs that analyzes a user's skill profile from the database and recommends personalized real-world projects to build.
- Developed a real-time anti-cheating proctoring system combining YOLOv8n (multi-face and phone detection) with MediaPipe (gaze tracking, head pose, posture analysis) running concurrently on live webcam streams.

AI Stylist – Multi-Agent Assistant for a High-End Fashion Brand *LangGraph, LangChain, Playwright, PostgreSQL, ORM, Python*

- Designed the relational schema behind the LangGraph-orchestrated agent flow, modelling the product catalog and conversation state so both agents could query it directly.
- Wrote the ORM layer that exposed the schema to the agents as structured query interfaces instead of raw SQL, keeping data access consistent across agent flows.
- Scraped the client's full website with Playwright and built the cleaning and normalization pass that turned raw pages into the structured dataset the agents retrieve over.

Multimodal RAG – Document QA System *VLM, PyMuPDF4LLM, fitz, Qdrant, hybrid retrieval*

- Built a multimodal RAG pipeline retrieving over both text and images, using a vision-language model (VLM) to interpret figures alongside font-aware text extracted via PyMuPDF4LLM and embedded images via fitz.
- Implemented recursive chunking and hybrid retrieval (dense + sparse embeddings) on Qdrant to improve recall and ranking over dense-only search.

LLaMA 3.1 8B Fine-Tuning – Text-to-SQL *QLoRA, bitsandbytes, PEFT, HuggingFace SFTTrainer, MLflow*

- Fine-tuned LLaMA 3.1 8B on the Spider Text-to-SQL dataset using QLoRA (4-bit quantization) on Kaggle GPUs.
- Applied PEFT via LoRA adapters for parameter-efficient fine-tuning, tracking experiments and metrics with MLflow.

Technical Skills

LLM & GenAI: LLaMA, QLoRA, PEFT, bitsandbytes, HuggingFace Transformers, SFTTrainer, LangGraph

RAG & Vector DBs: Qdrant, PG-Vector, hybrid retrieval, reranking, pdfplumber, PyMuPDF4LLM, chunking strategies

ML/DL Frameworks: PyTorch, NumPy, Pandas, HuggingFace

MLOps & Deployment: MLflow, Docker, FastAPI, AWS Lambda

Frontend & DB: React, Next.js, PostgreSQL, Mongo

Programming: Python, C++, GitHub

Education

B.S. Computer Science

2022 – 2026

FAST National University of Computer and Emerging Sciences (NUCES)